DG INTERNS HUB

Cybersecurity Internship

WEEK 1 ASSIGNMENT

Networking Fundamentals

Submitted by: Amar Pratap

Internship: Cybersecurity Internship

Week: 1

Submission Date: 7 September 2026

Objective: Build a strong foundation in networking and understand how real-world systems communicate.

1. Introduction to Networking

Computer networking connects computers, mobile devices, servers, routers, switches and other systems so that they can exchange data and share resources. Networking is a core foundation of cybersecurity because security professionals need to understand addressing, protocols, routing, traffic flow and communication between systems.

Week 1 Objectives

• Understand the seven layers of the OSI model.

• Understand the four layers of the TCP/IP model and compare it with OSI.

• Study HTTP/HTTPS, FTP, DNS and DHCP.

• Learn IPv4 addressing, public/private addresses and basic subnetting.

• Use basic Windows network commands.

• Build and configure a PC → Switch → Router topology in Cisco Packet Tracer.

Topics Covered

Area	Work Covered
Theory	OSI, TCP/IP, protocols, IP addressing and subnetting
Command-line practice	ping, ipconfig, tracert
Simulation	PC → 2960 Switch → Cisco 1941 Router
Addressing	PC 192.168.1.10/24; Router 192.168.1.1/24
Verification	Router interface configured and enabled with no shutdown

2. OSI Model

OSI stands for Open Systems Interconnection. It is a seven-layer reference model used to understand how data moves between networked systems.

Layer	Name	Main Function	Examples	Real-life example
7	Application	Provides network services to user applications.	HTTP/HTTPS, FTP, DNS	Browser accessing a website
6	Presentation	Handles data translation, encryption/decryption and compression.	Data representation, encryption	Protected web communication
5	Session	Establishes, manages and terminates communication sessions.	Session management	Logged-in application session
4	Transport	Provides end-to-end delivery, segmentation, flow and error control.	TCP, UDP	Reliable file transfer
3	Network	Provides logical addressing and routing between networks.	IP	Router forwarding packets
2	Data Link	Uses frames and MAC addresses for local-link delivery.	Ethernet	Switch forwarding frames
1	Physical	Transmits raw bits through physical media or signals.	Cables, radio, signals	Ethernet cable / Wi-Fi signal

Memory trick: A P S T N D P — All People Seem To Need Data Processing.

Important: OSI is primarily a conceptual/reference model used for learning, design and troubleshooting.

3. TCP/IP Model

TCP/IP (Transmission Control Protocol/Internet Protocol) is the practical protocol architecture used for communication across the Internet and modern IP networks.

Layer	Function	Examples	Real-life example
Application	Services used by network applications.	HTTP, HTTPS, FTP, DNS, SMTP	Browser, email and file transfer
Transport	Process-to-process communication.	TCP, UDP	Reliable or low-overhead delivery
Internet	Logical addressing and routing.	IP, ICMP	Packet routing between networks
Network Access	Local network delivery and physical transmission.	Ethernet, Wi-Fi	LAN communication

How data moves

When an application sends data, information moves down the TCP/IP layers before transmission. At the destination, the process is reversed and the receiving application gets the data.

Layer order: Application → Transport → Internet → Network Access

4. OSI vs TCP/IP — Detailed Comparison

The two models describe network communication at different levels of abstraction. OSI separates functions into seven layers, while TCP/IP groups several functions into four broader layers.

Point	OSI Model	TCP/IP Model
Purpose	Reference model used to explain and standardize networking functions.	Practical protocol architecture used by the Internet and IP networks.
Number of layers	7 layers.	4 layers.
Application area	Application, Presentation and Session are separate layers.	These three OSI functions are grouped into one Application layer.
Transport	Transport layer provides end-to-end communication.	Transport layer provides process-to-process communication using TCP/UDP.
Network area	Network layer handles logical addressing and routing.	Internet layer handles IP addressing and routing.
Lower layers	Data Link and Physical are separate layers.	Network Access combines local-link and physical transmission functions.
Model type	Conceptual/reference model.	Practical architecture/protocol suite.
Typical use	Teaching, design, troubleshooting and explaining responsibilities.	Real-world Internet and IP networking.

Layer Mapping

OSI	TCP/IP	What happens
Application + Presentation + Session	Application	Application services, data representation and session-related functions.
Transport	Transport	End-to-end/process-to-process communication.
Network	Internet	IP addressing and routing.
Data Link + Physical	Network Access	Frames/local delivery plus physical transmission.

5. Network Protocols

Protocols are agreed rules that define how devices communicate and exchange data.

Protocol	Full Form	What it does	Where used
HTTP	HyperText Transfer Protocol	Transfers web resources.	Web browsing
HTTPS	HTTP Secure	Provides secure web communication using TLS.	Logins, banking, shopping and websites
FTP	File Transfer Protocol	Transfers files between client and server.	File upload/download
DNS	Domain Name System	Resolves domain names to IP addresses.	Finding a server address
DHCP	Dynamic Host Configuration Protocol	Automatically provides network configuration.	Home/office Wi-Fi and LANs

HTTP vs HTTPS

HTTP uses port 80 by default. HTTPS commonly uses port 443 and protects web communication using TLS.

DNS example

A user enters a domain name such as **google.com**. DNS helps the device discover an IP address associated with that name.

DHCP example

When a device joins a LAN/Wi-Fi network, a DHCP server can automatically provide an IP address, subnet mask, default gateway and DNS information.

6. IP Addressing and Basic Subnetting

An IP address is a logical address used to identify a network interface for IP communication. IPv4 addresses contain 32 bits and are normally written as four decimal octets.

Example: 192.168.1.10

Private IPv4 ranges

• 10.0.0.0/8

• 172.16.0.0/12

• 192.168.0.0/16

Public vs Private IP

Type	Description
Public IP	Used for Internet-facing communication; normally allocated by an ISP/provider.
Private IP	Used inside private networks and is not directly routed across the public Internet.

Basic subnetting example: 192.168.1.0/24

Item	Value
CIDR prefix	/24
Subnet mask	255.255.255.0
Total IPv4 addresses	256
Network address	192.168.1.0
Usable host range	192.168.1.1 – 192.168.1.254
Broadcast address	192.168.1.255
Usable host addresses	254

Why /24? In a /24 network, 24 bits identify the network and 8 bits remain for host addressing. Two addresses are reserved for network and broadcast, leaving 254 usable host addresses.

7. Practical Task — Windows Network Commands

The following commands were tested in Windows PowerShell. The screenshots below are the user's practical evidence.

7.1 ping google.com

The **ping** command tests reachability and reports round-trip response times. The supplied screenshot shows 4 packets sent, 4 received and 0% packet loss.

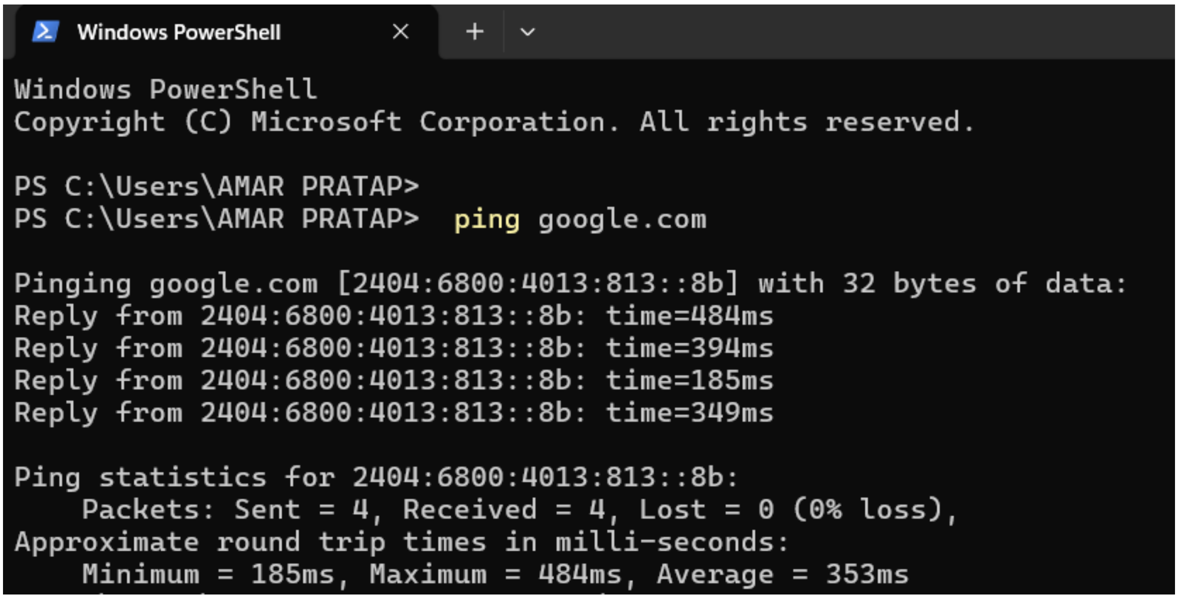


Figure 1: User's ping test to google.com showing 0% packet loss.

7.2 ipconfig — Correct Windows Syntax

The correct Windows command is **ipconfig** by itself. The earlier screenshot shows **ipconfig / ifconfig**, which is invalid command syntax on Windows. This is a command-entry issue, not evidence of a network failure.

`Correct command:` `ipconfig`

`Useful variants:` `ipconfig /all | ipconfig /release | ipconfig /renew`

For final evidence, capture a clean output of **ipconfig** or **ipconfig /all** showing the IPv4 address, subnet mask and default gateway. The original error screenshot is not presented as successful output.

8. tracert and Cisco Packet Tracer Setup

8.1 tracert google.com

The **tracert** command traces the path from the computer toward a destination. The supplied screenshot shows multiple hops and several 'Request timed out' entries. Intermediate devices may not return traceroute probes, so a timeout does not automatically mean the overall route failed.

```
                                          compartments
PS C:\Users\AMAR PRATAP> tracert google.com

Tracing route to google.com [2404:6800:4013:813::8b]
over a maximum of 30 hops:

  1     4 ms     2 ms     3 ms  2409:40d1:2004:9551::f9
  2     *        *        *     Request timed out.
  3   979 ms   576 ms     *     2405:200:520b:25:3925::1
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    83 ms    77 ms    81 ms  2405:200:801:1500::7a6
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9   280 ms   171 ms   138 ms  2405:200:802:760::8
 10   267 ms   177 ms   245 ms  2405:200:802:760::8
 11     *        *        *     Request timed out.
 12   947 ms   831 ms  1112 ms  2404:6800:8285:c0::1
 13   685 ms   562 ms   877 ms  2404:6800:8285:c0::1
 14  1099 ms   789 ms   694 ms  2001:4860:0:1::442
 15   594 ms   811 ms   654 ms  2001:4860:0:1::a7ec
 16   743 ms   739 ms   617 ms  2001:4860::c:4004:53bf
 17  1390 ms   845 ms   653 ms  2001:4860::c:4004:2137
 18   690 ms   140 ms   129 ms  2001:4860:0:1::2496
 19     *        *        *     Request timed out.
 20     *        *        *     Request timed out.
 21     *        *        *     Request timed out.
 22     *        *        *     Request timed out.
 23     *        *        *     Request timed out.
 24     *        *        *     Request timed out.
 25     *        *        *     Request timed out.
 26     *        *        *     Request timed out.
 27   144 ms   230 ms   347 ms  lcdels-in-f139.1e100.net [2404:6800:4013:813::8b]

Trace complete.
```

Figure 2: User's tracert google.com output.

8.2 Cisco Packet Tracer

The required topology is **PC** → **Switch** → **Router** using Copper Straight-Through cables. The user's topology uses a PC-PT, a 2960-24TT switch and a Cisco 1941 router.

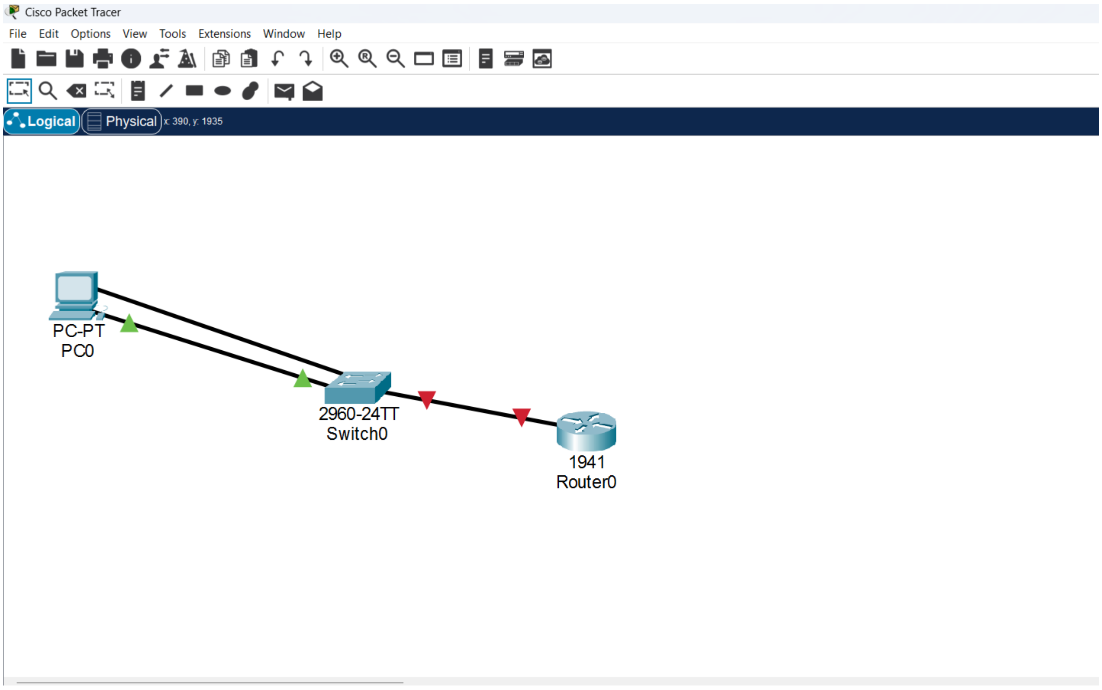


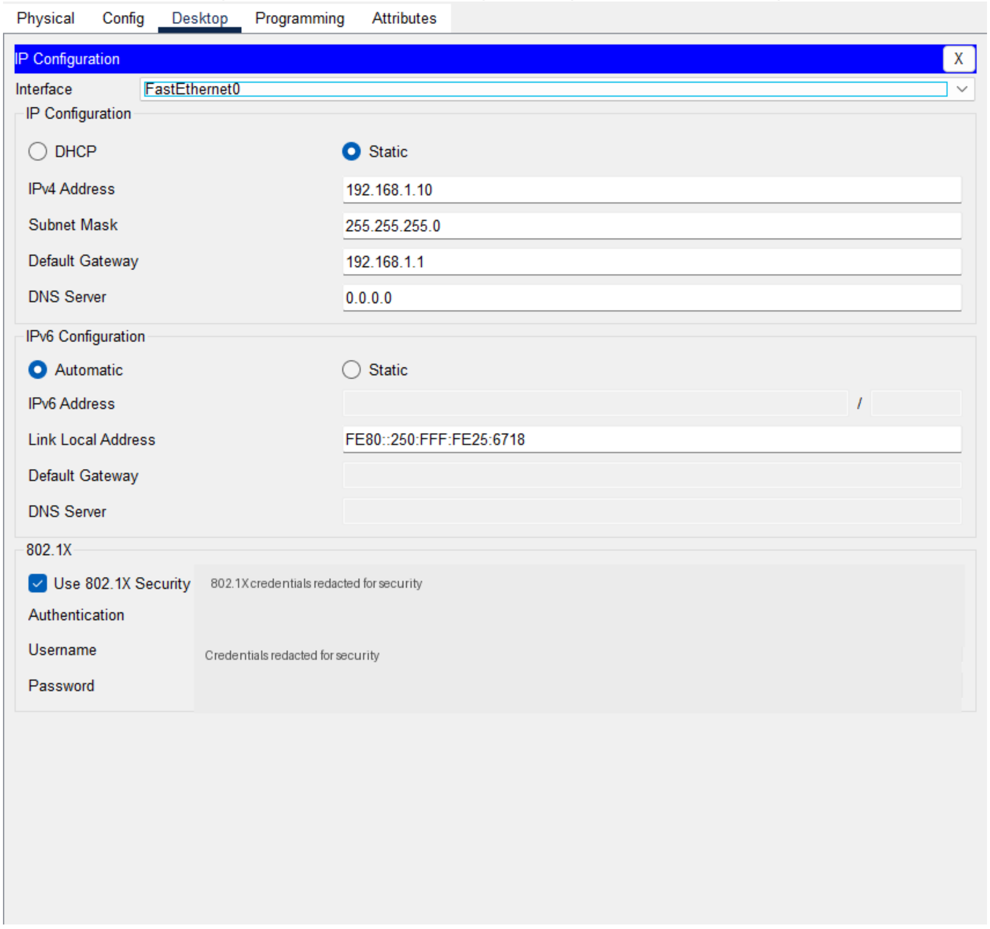
Figure 3: User's Cisco Packet Tracer PC → Switch → Router topology.

9. Packet Tracer IP Configuration

The PC was configured with the assignment's specified IPv4 settings.

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
Router0 (G0/0)	192.168.1.1	255.255.255.0	—

Figure 4: User's PC IP configuration (credentials redacted).



Router CLI configuration used

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

The router screenshot shows the interface state changing to **up** after **no shutdown**.

10. Router Evidence, Learning Outcomes & Final Checklist

10.1 Router Configuration Evidence

The supplied Cisco 1941 router screenshot shows GigabitEthernet0/0 configured with 192.168.1.1/24. The **no shutdown** command was entered and the interface reported a state change to up.

```
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5: User's Cisco 1941 router CLI configuration.

10.2 Learning Outcomes

• Understood the OSI model and its seven layers.

• Understood the TCP/IP model and its relationship to OSI.

• Understood the OSI-to-TCP/IP layer mapping in detail.

• Learned HTTP/HTTPS, FTP, DNS and DHCP.

• Learned IPv4 addressing, public/private IP concepts and /24 subnetting.

• Used ping and tracert for basic network testing.

• Corrected the Windows ipconfig command syntax.

• Built a PC → Switch → Router topology in Cisco Packet Tracer.

• Configured the PC and router with the assignment's IPv4 addressing.

Conclusion

This Week 1 assignment provided a practical foundation in networking. The theory explains how communication is organized into layers and protocols, while the practical work demonstrates IP addressing, router configuration and basic troubleshooting. These concepts are essential for cybersecurity because network traffic, addressing and protocols are involved in defensive monitoring and security investigations.

Final evidence checklist before submission

1. Run **ipconfig** by itself and capture a clean screenshot showing IPv4 address, subnet mask and default gateway.

2. In Cisco Packet Tracer, verify connectivity from PC0 using **ping 192.168.1.1** and capture the successful result.

Note: These two screenshots are not fabricated in this corrected report. They still need to be captured from the user's own system if the internship requires final proof.

DG Interns Hub

Week 1: Networking Fundamentals

Cybersecurity Internship
Submitted by: Amar Pratap

OSI Model

Seven-layer reference model

7 Application — network services to applications

6 Presentation — translation, encryption and compression

5 Session — establishes and manages sessions

4 Transport — end-to-end delivery (TCP/UDP)

3 Network — IP addressing and routing

2 Data Link — frames and MAC addresses

1 Physical — bits, signals and media

TCP/IP Model

Four-layer practical protocol architecture

- Application — HTTP, HTTPS, FTP, DNS and other application protocols
- Transport — TCP and UDP
- Internet — IP and routing
- Network Access — Ethernet, Wi-Fi and local/physical delivery
- OSI mapping: Application/Presentation/Session → Application; Transport → Transport; Network → Internet; Data Link/Physical → Network Access

Network Protocols

HTTP — web communication

HTTPS — secure web communication using TLS

FTP — file transfer between client and server

DNS — resolves domain names to IP addresses

DHCP — automatically provides IP configuration

IP Addressing & Subnetting

IPv4 uses 32 bits and is written as four decimal octets

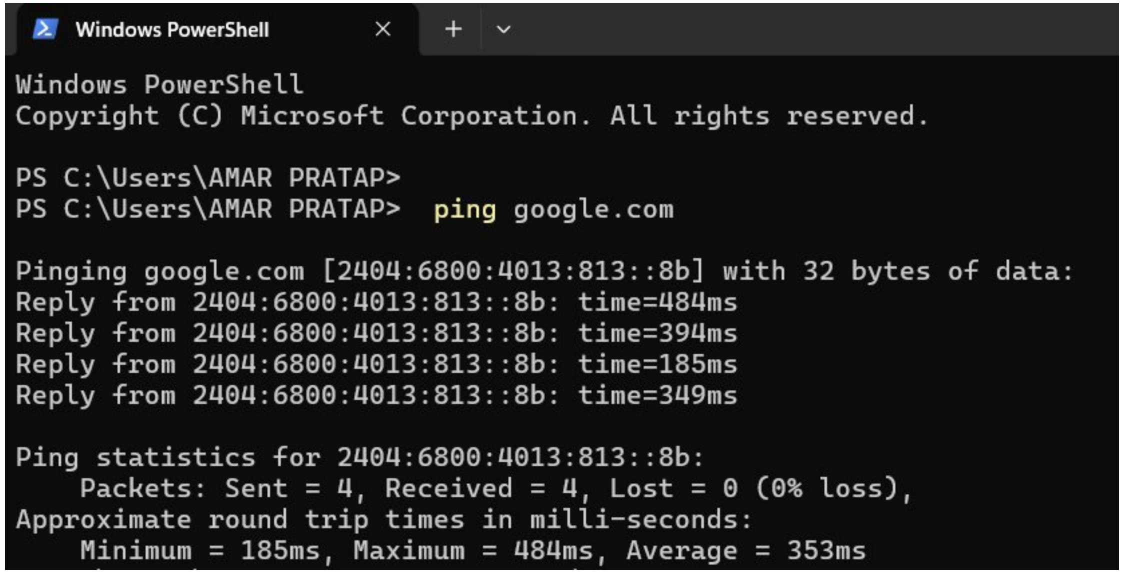
Private ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

Public IPs are used for Internet-facing communication

Example subnet: 192.168.1.0/24

Network: 192.168.1.0 | Usable hosts: 192.168.1.1–192.168.1.254 | Broadcast: 192.168.1.255

Practical: Windows Network Commands

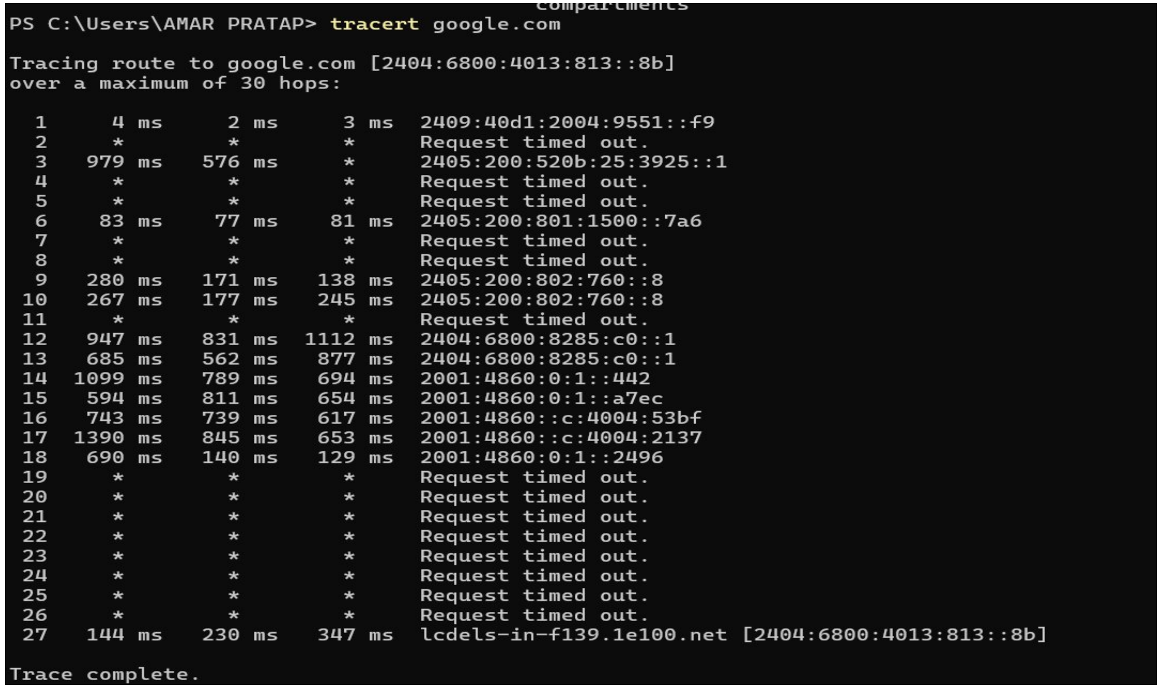


```
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\AMAR PRATAP>
PS C:\Users\AMAR PRATAP>  ping google.com

Pinging google.com [2404:6800:4013:813::8b] with 32 bytes of data:
Reply from 2404:6800:4013:813::8b: time=484ms
Reply from 2404:6800:4013:813::8b: time=394ms
Reply from 2404:6800:4013:813::8b: time=185ms
Reply from 2404:6800:4013:813::8b: time=349ms

Ping statistics for 2404:6800:4013:813::8b:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 185ms, Maximum = 484ms, Average = 353ms
```



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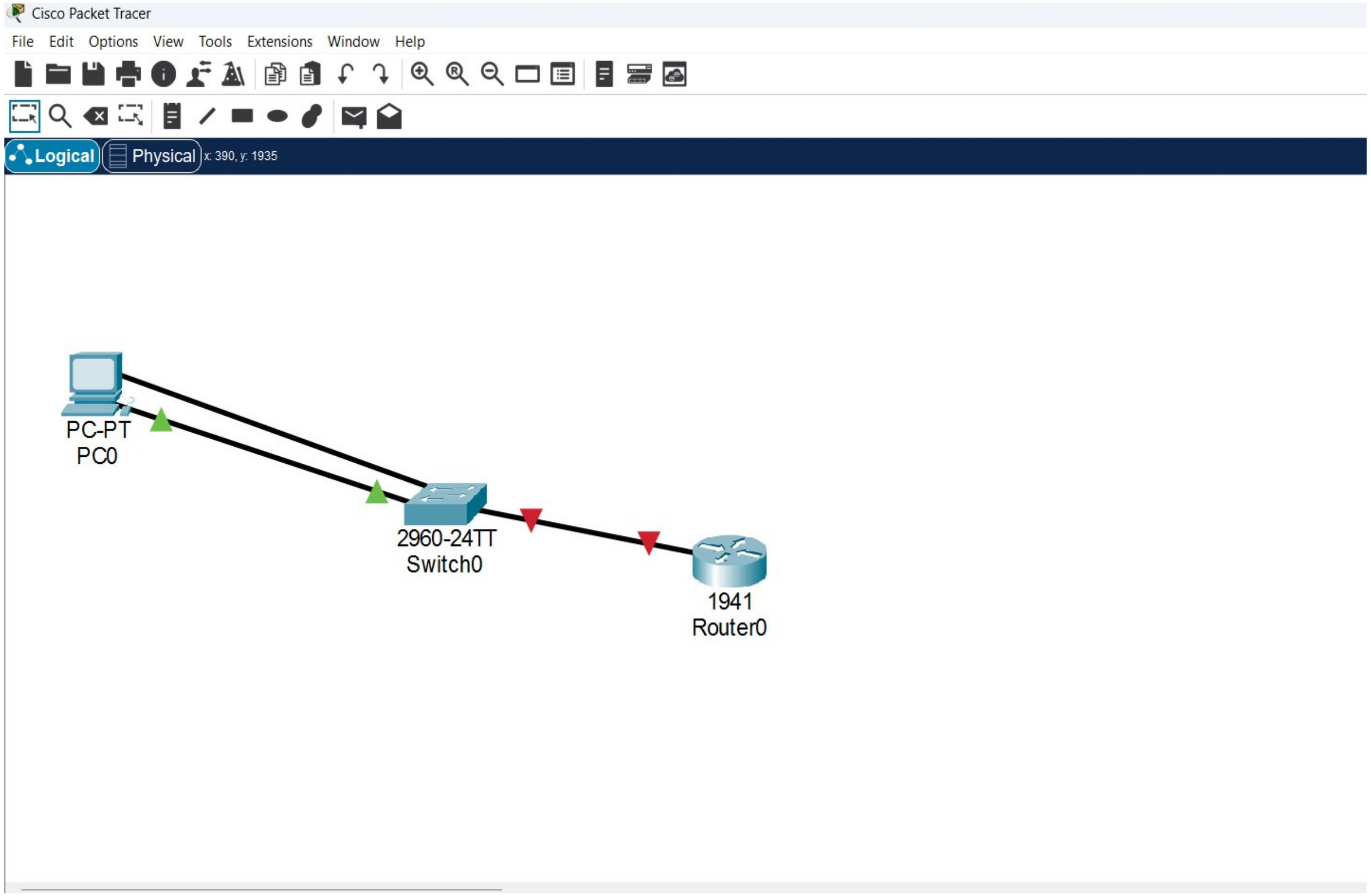
Trace complete.
```

ping google.com → 4 replies, 0% packet loss in the supplied evidence

tracert google.com → multiple hops; some intermediate timeouts are visible

ipconfig → run the command by itself on Windows

Practical: Cisco Packet Tracer



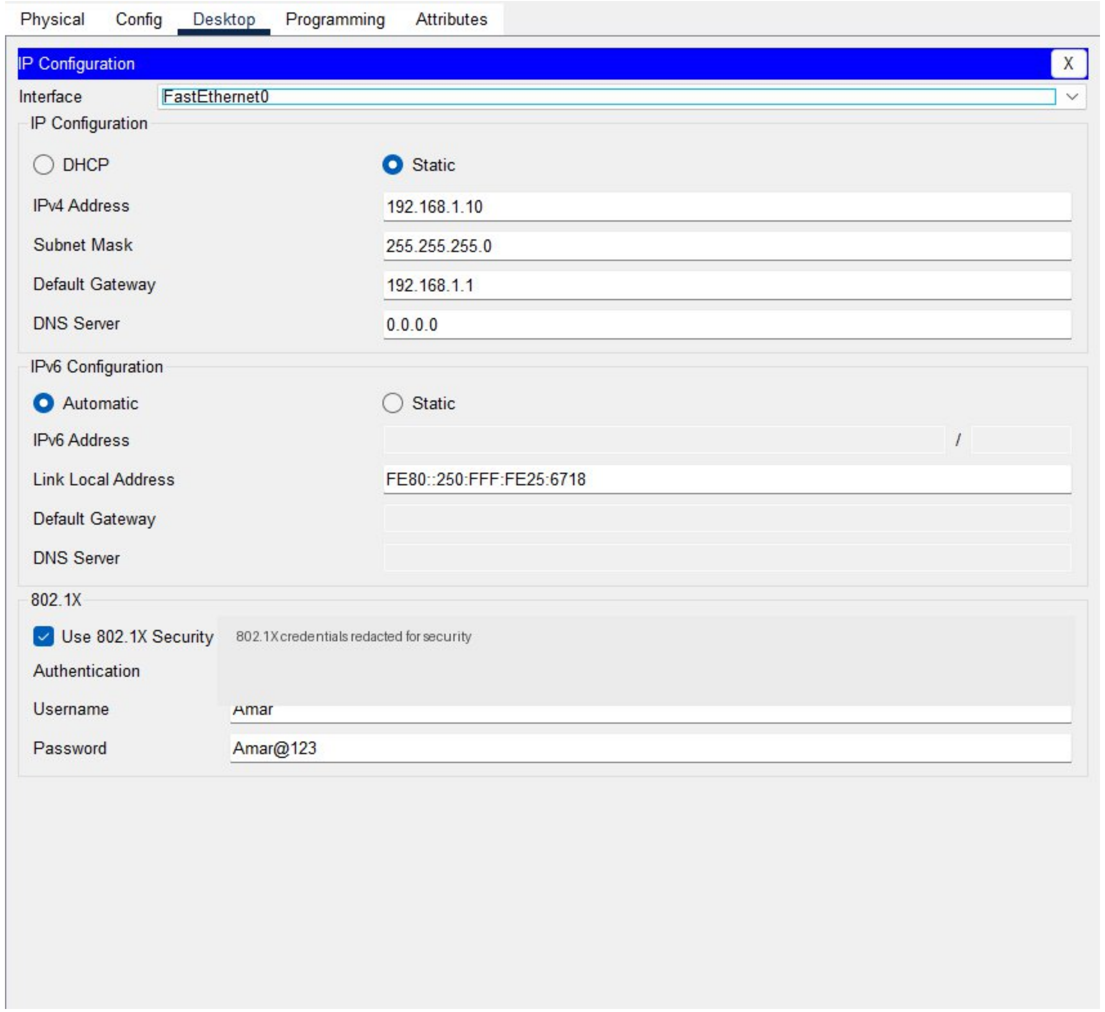
Topology: PC → 2960 Switch → Cisco 1941 Router

Connection: Copper Straight-Through

PC: 192.168.1.10/24

Router G0/0: 192.168.1.1/24

Configuration & Learnings

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http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

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%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

PC and router IP addresses configured

Router interface enabled with no shutdown

Networking concepts practiced for cybersecurity